

Cluster Timing (Part 2)

Dean Reiter

DarkLight Software Meeting

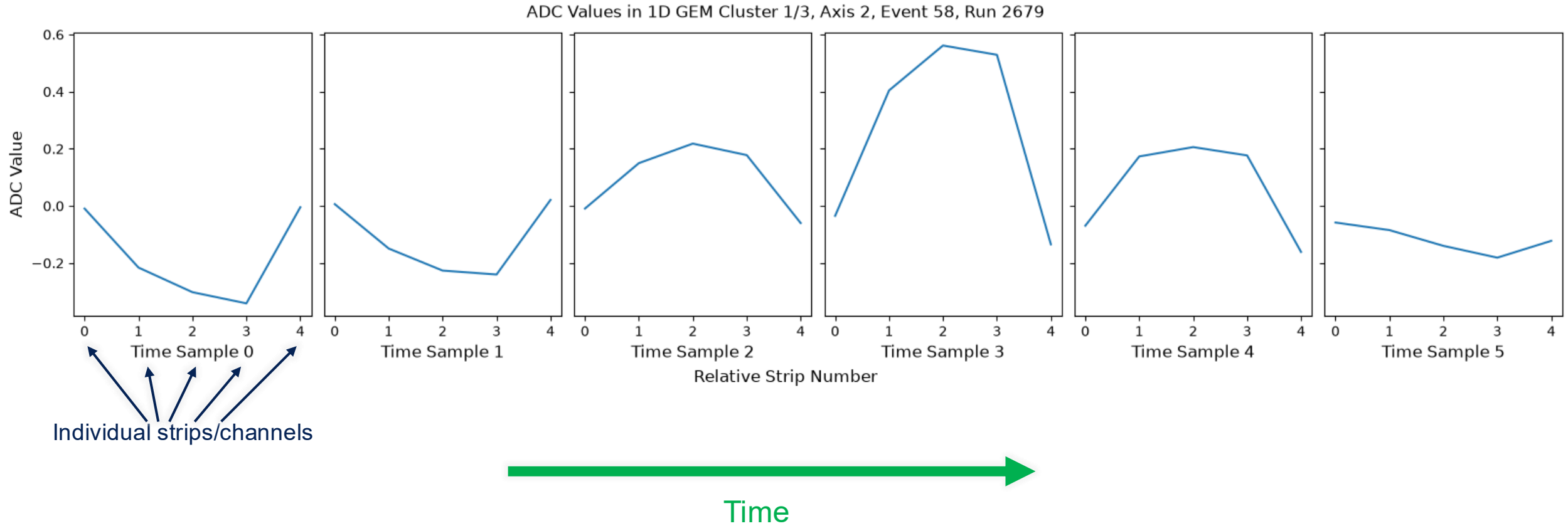
July 13, 2026

Example of a cluster

1

Pedestal-subtracted values across five strips over six time samples

These five strips were identified as containing a cluster in this event

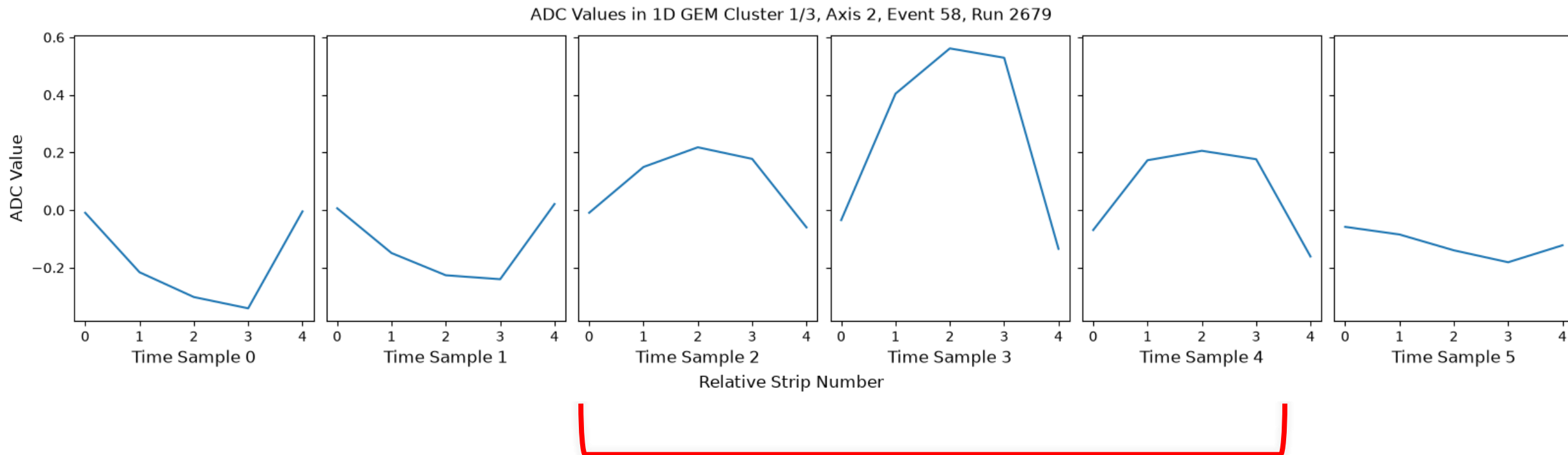


Example of a cluster

2

Pedestal-subtracted values across five strips over six time samples

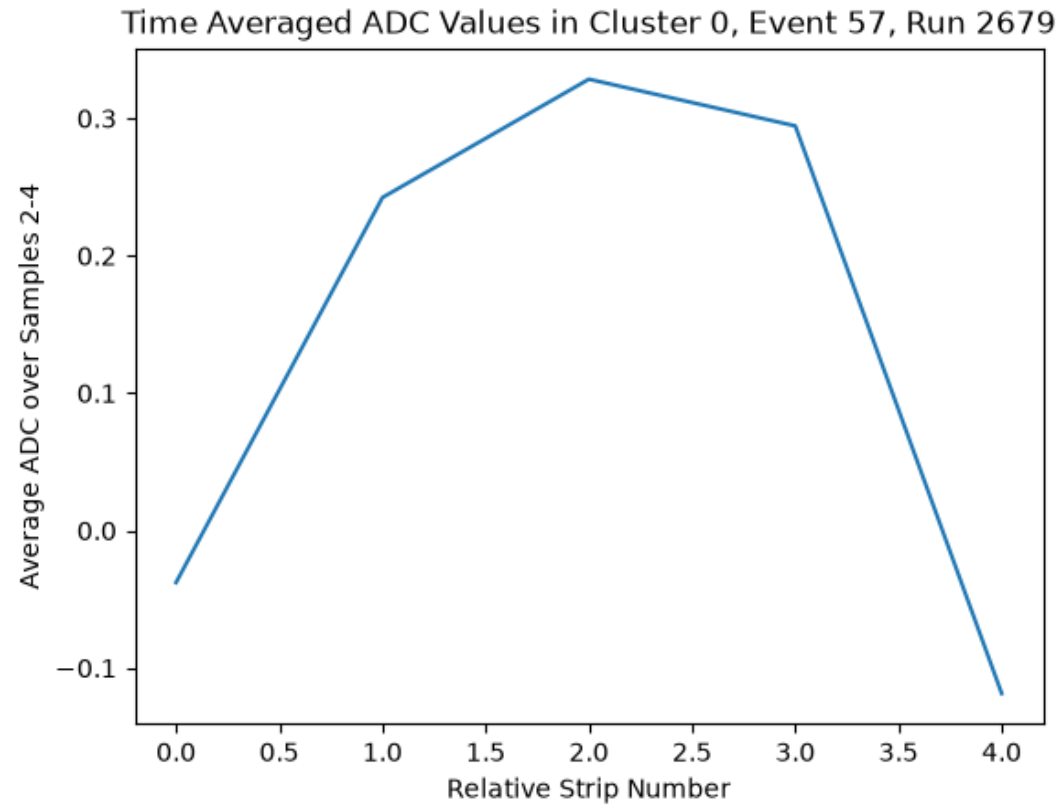
These five strips were identified as containing a cluster in this event



The code takes the average of samples 2 through 4 before searching for clusters

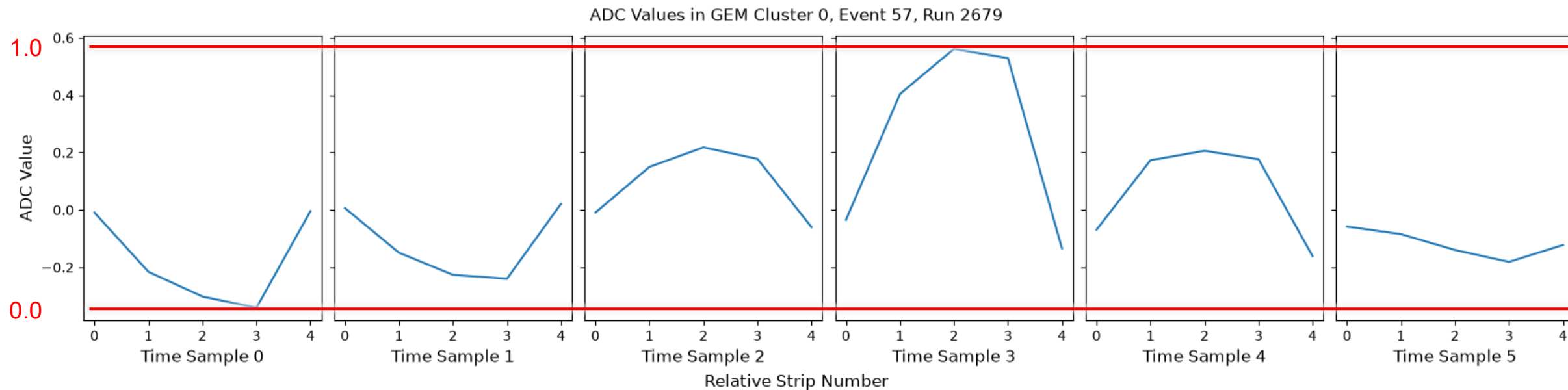
Example of a cluster

This is the time-averaged data which Andy's algorithm identified as a cluster



Approximating Cluster Time

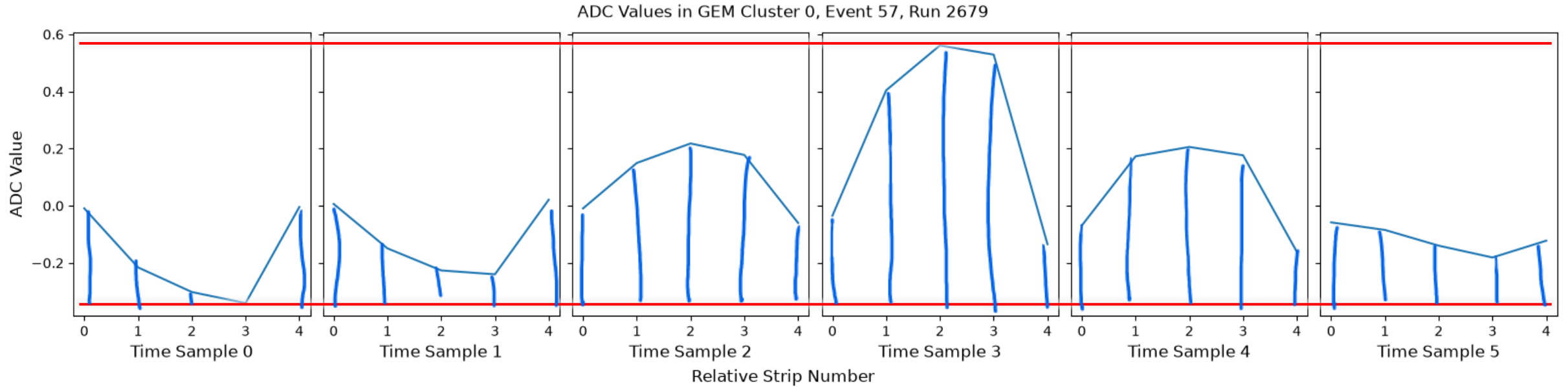
4



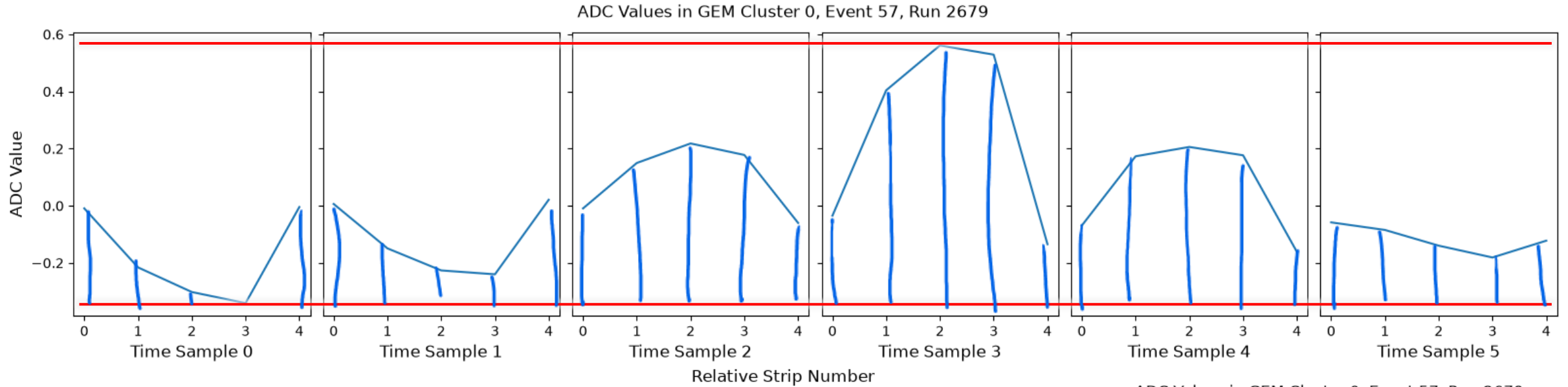
1. Normalize all values to the overall minimum and maximum

Approximating Cluster Time

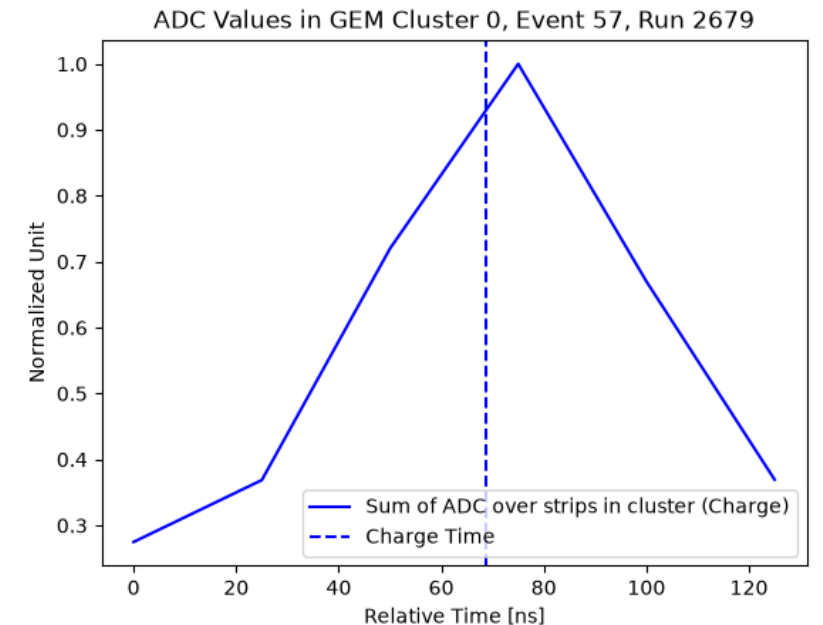
5



1. Normalize all values to the overall minimum and maximum
2. Calculate “charge” in each sample = sum of channel values



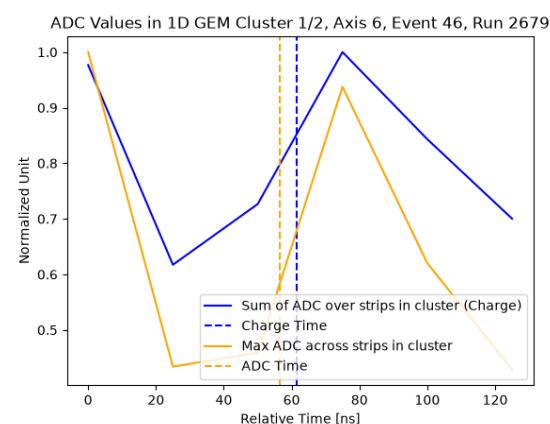
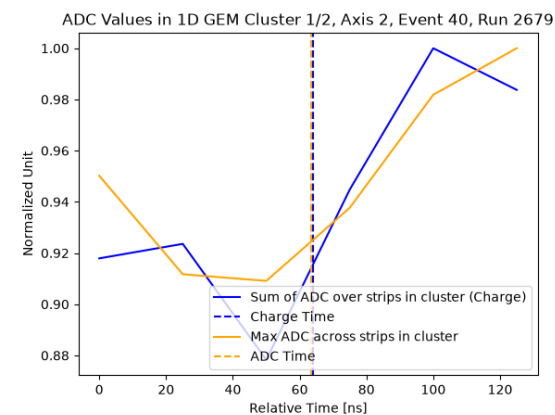
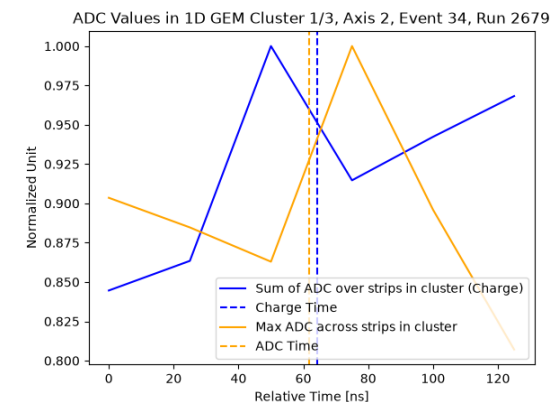
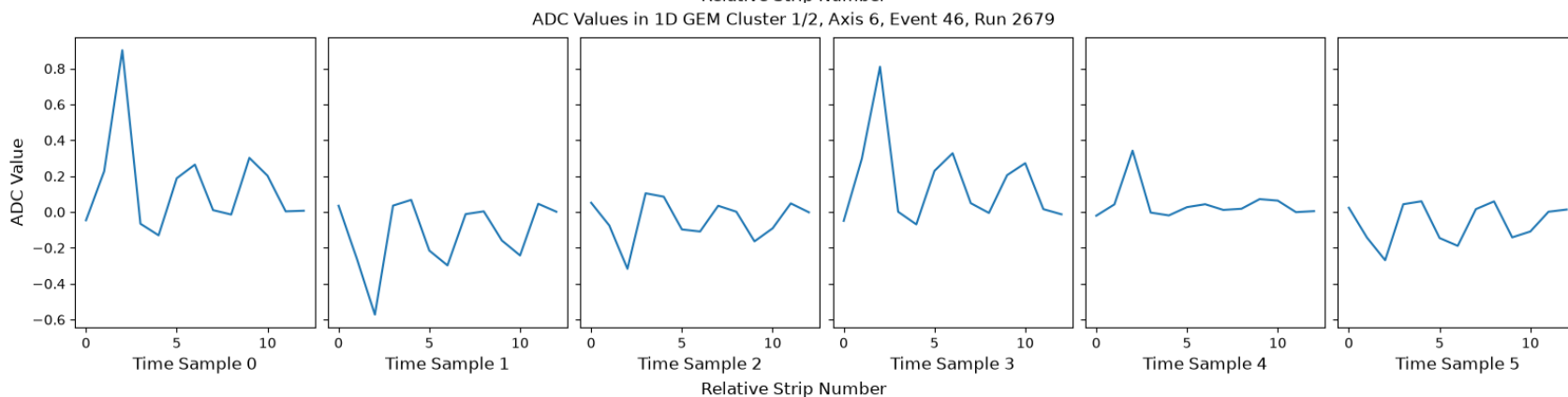
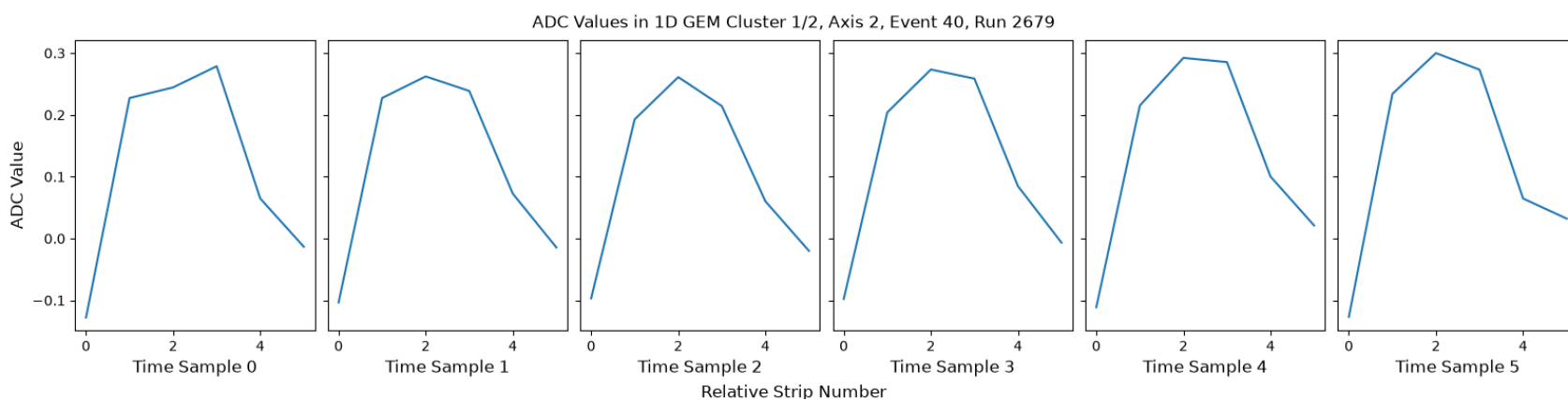
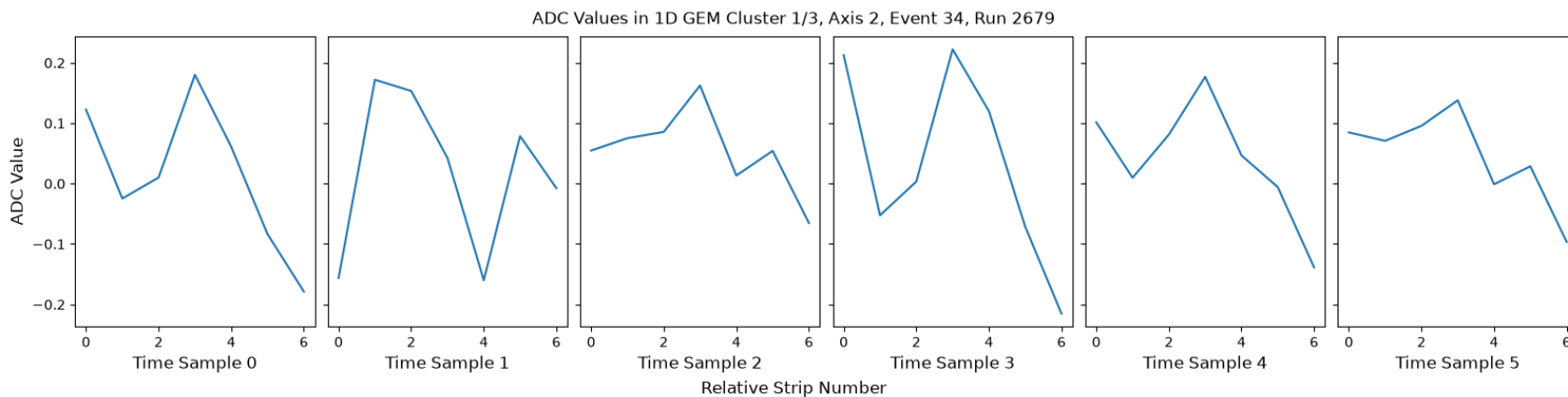
1. Normalize all values to the overall minimum and maximum
2. Calculate "charge" in each sample = sum of channel values
3. Calculate "weighted mean" of time values, where the weights are the charge values



- The slides I presented in the software meeting two weeks ago had errors
 - Uploaded revised slides to GitHub
- Method of determining cluster time is not very accurate or precise...
 - Averaging method might bias the time to the middle of the time window (62.5ns)
 - Can be improved
- Many clusters are not ideal
 - Low signal to noise ratio – especially in cosmic data
 - Peak detection algorithm misses many good clusters in elastic data

Variety of cluster signal shapes

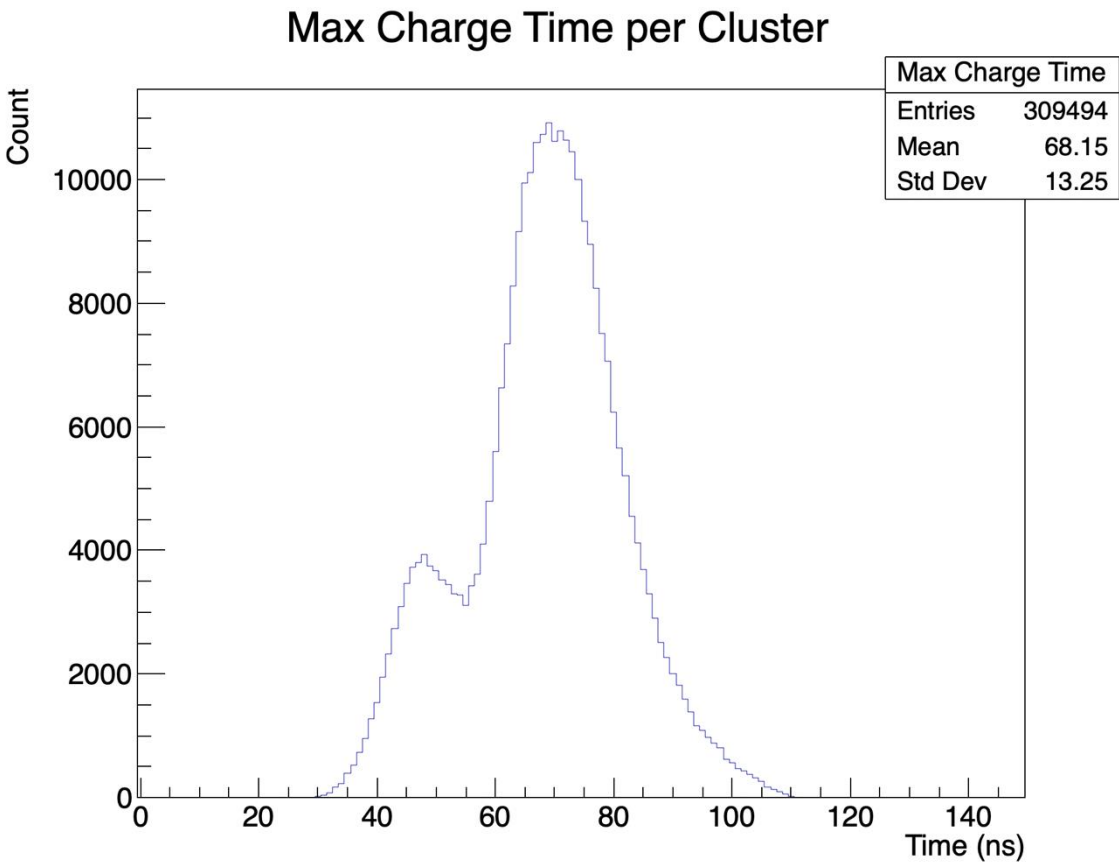
8



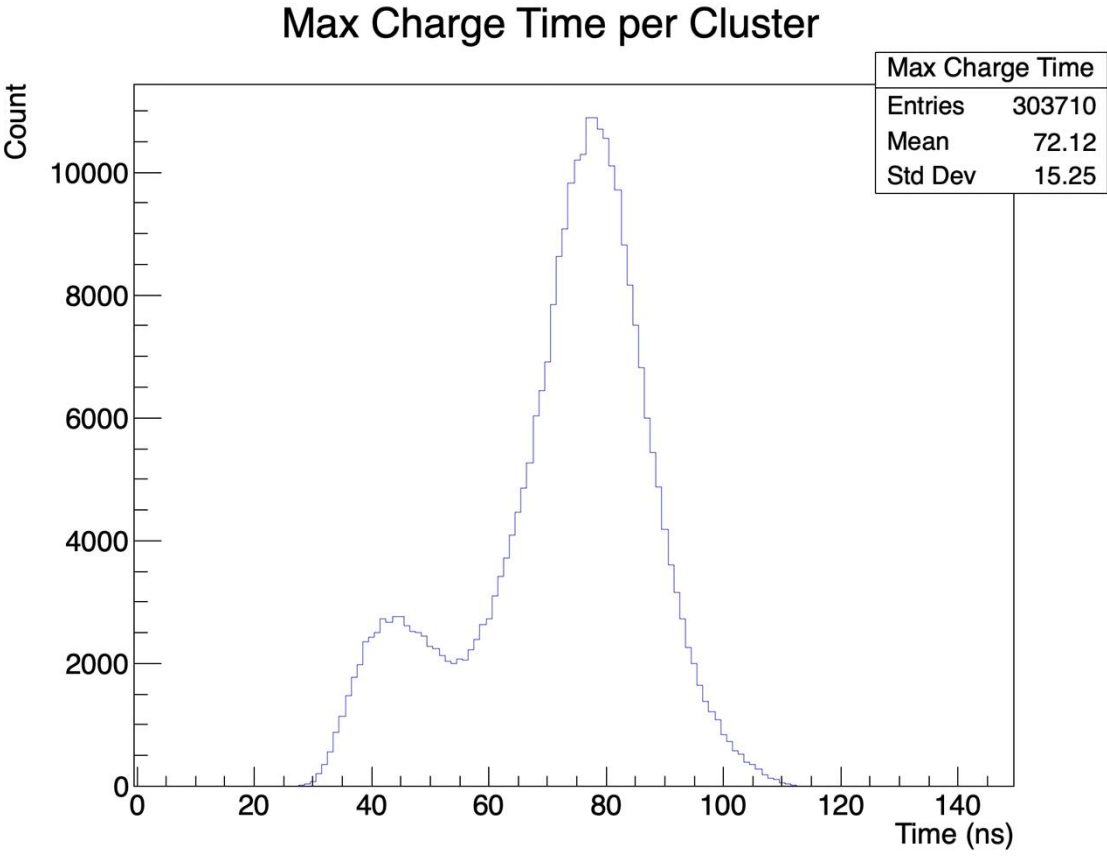
Charge Time

Left Top GEM clusters

Run 4010 - Elastic



X axis

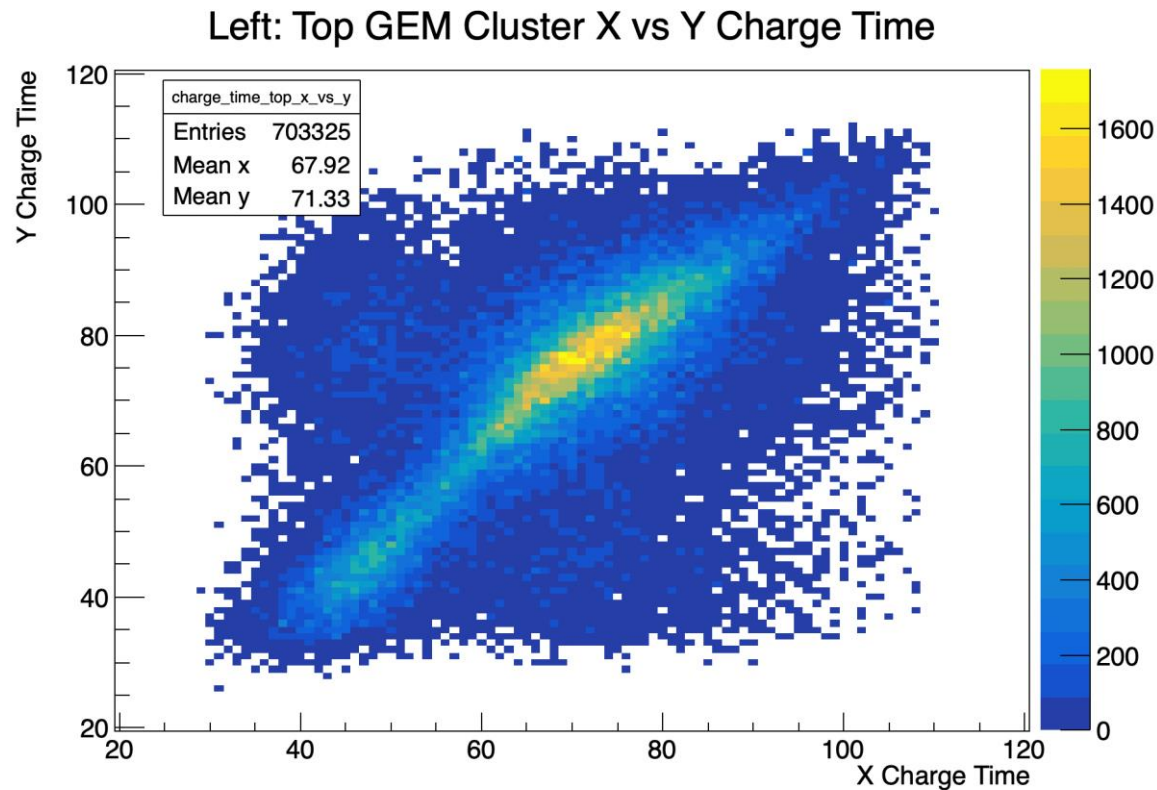


Y axis

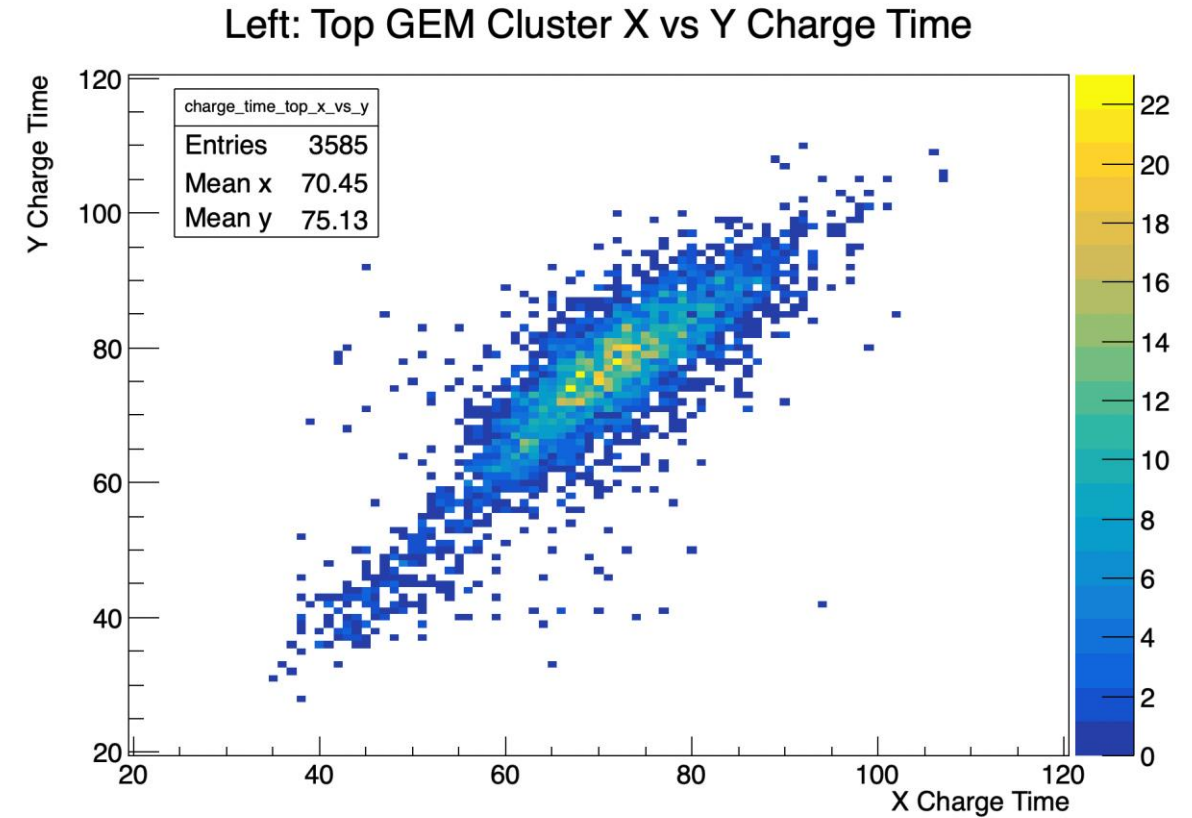
Charge Time

Run 4010 - Elastic

Left Top GEM clusters



All tracks



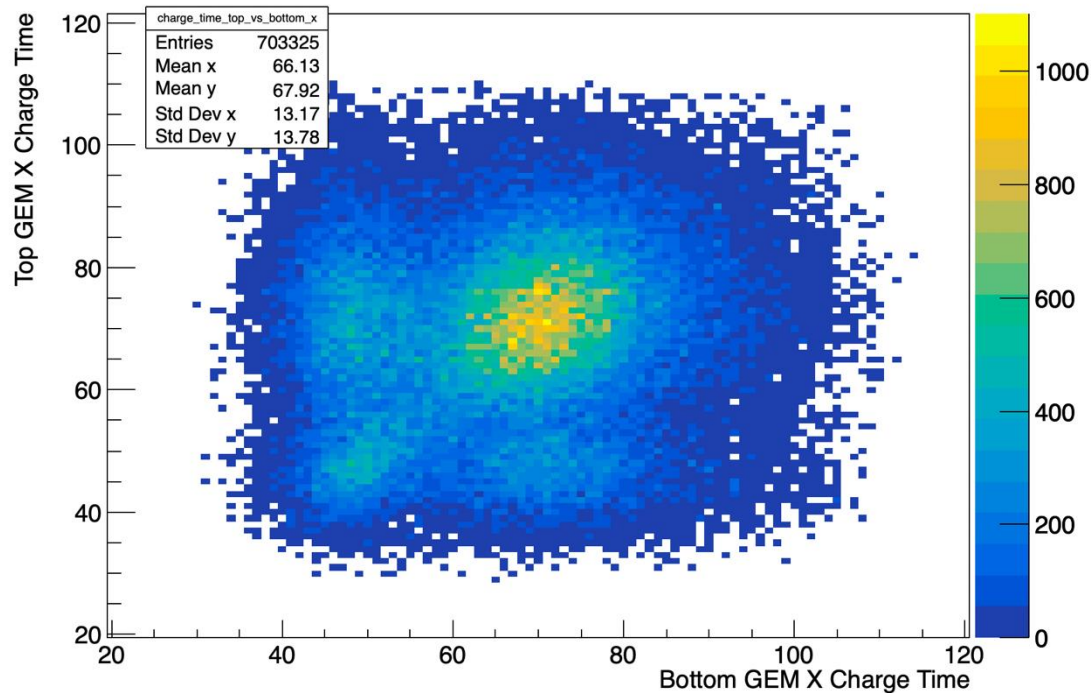
Single (one hit per detector per event) and
On-Paddle (hit scintillators) tracks

Charge Time

X axis clusters

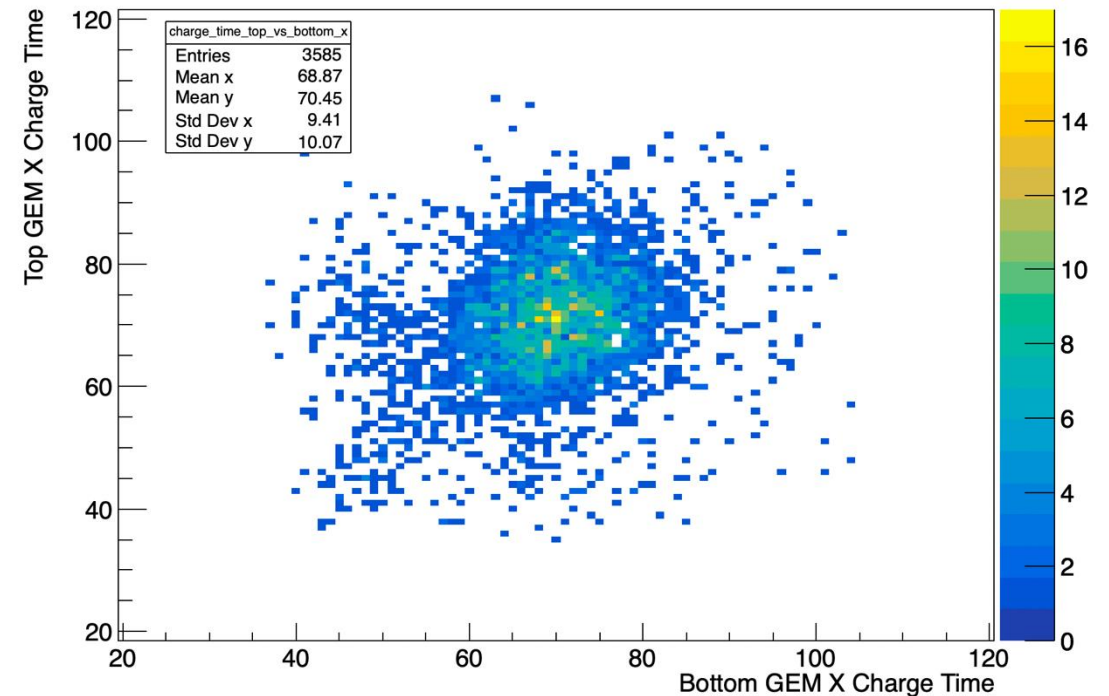
Run 4010 - Elastic

Left: Top vs Bottom GEM Cluster X Charge Time



All tracks

Left: Top vs Bottom GEM Cluster X Charge Time

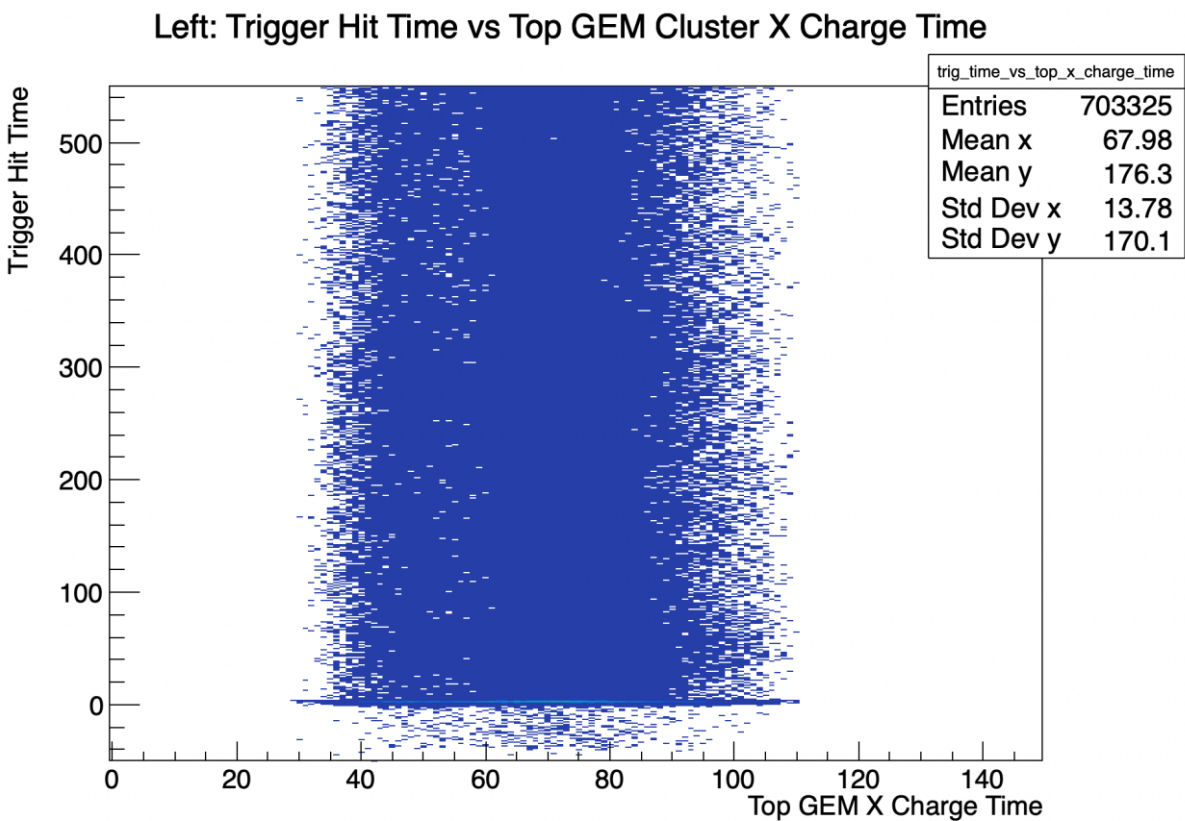


Single (one hit per detector per event) and
On-Paddle (hit scintillators) tracks

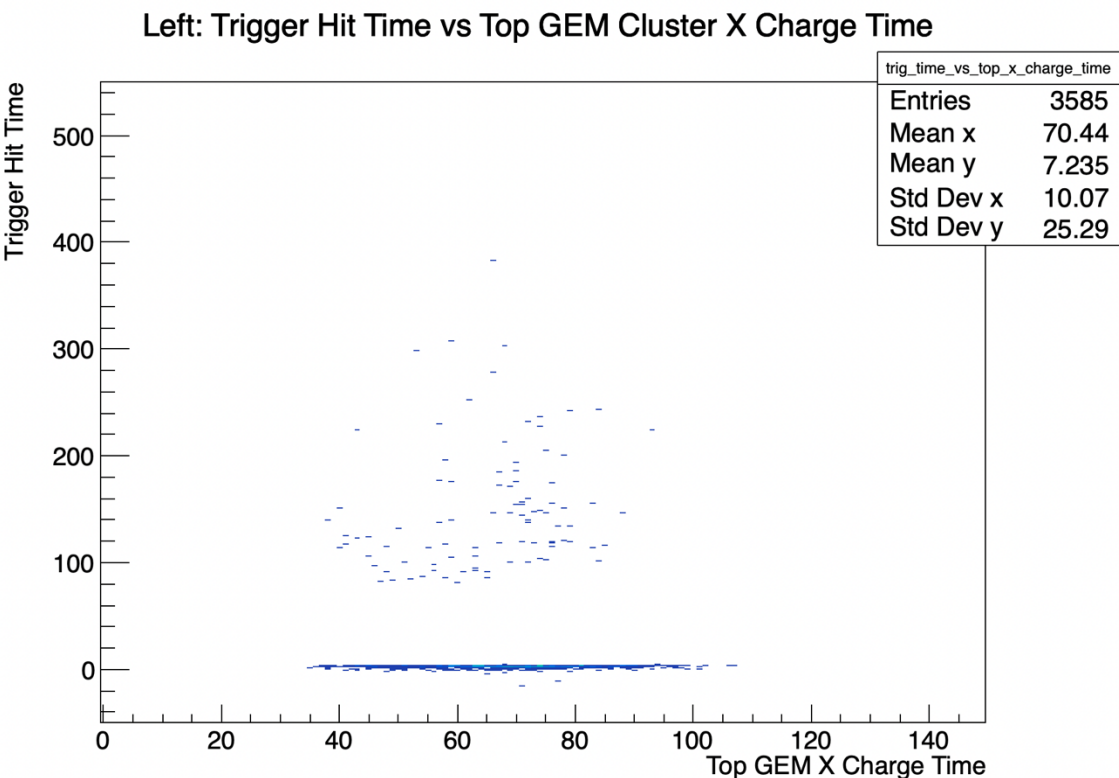
Charge Time

Trigger vs Left Top GEM X axis

Run 4010 - Elastic



All tracks



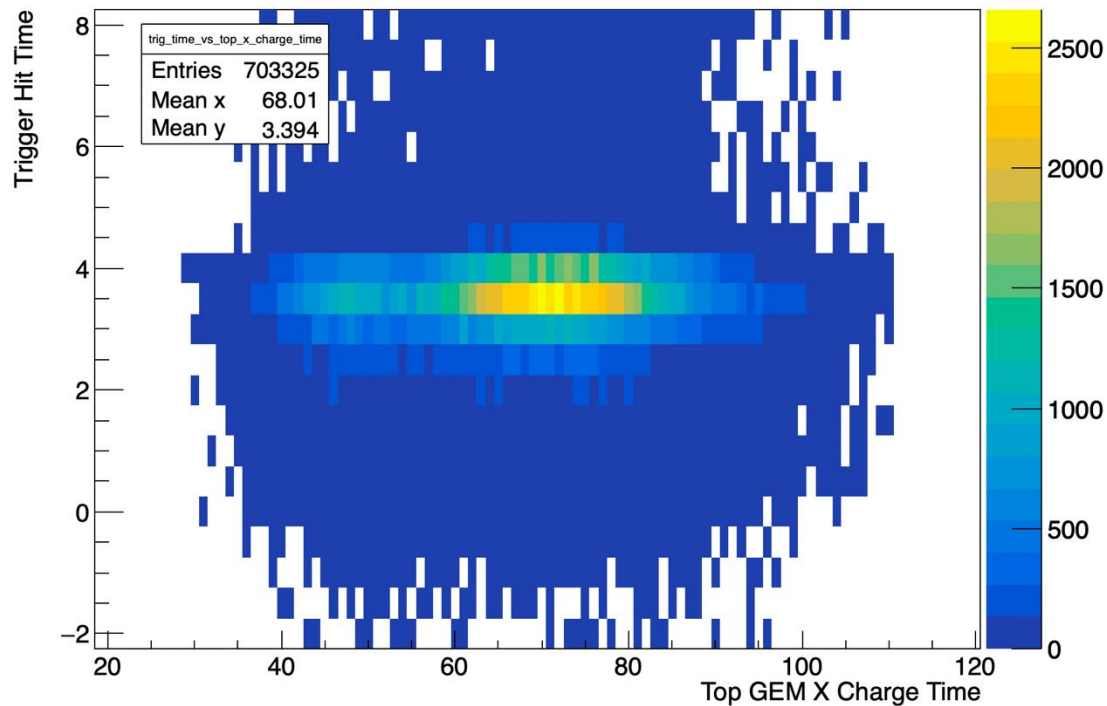
Single (one hit per detector per event) and On-Paddles (hits scintillators) tracks

Charge Time

Run 4010 - Elastic

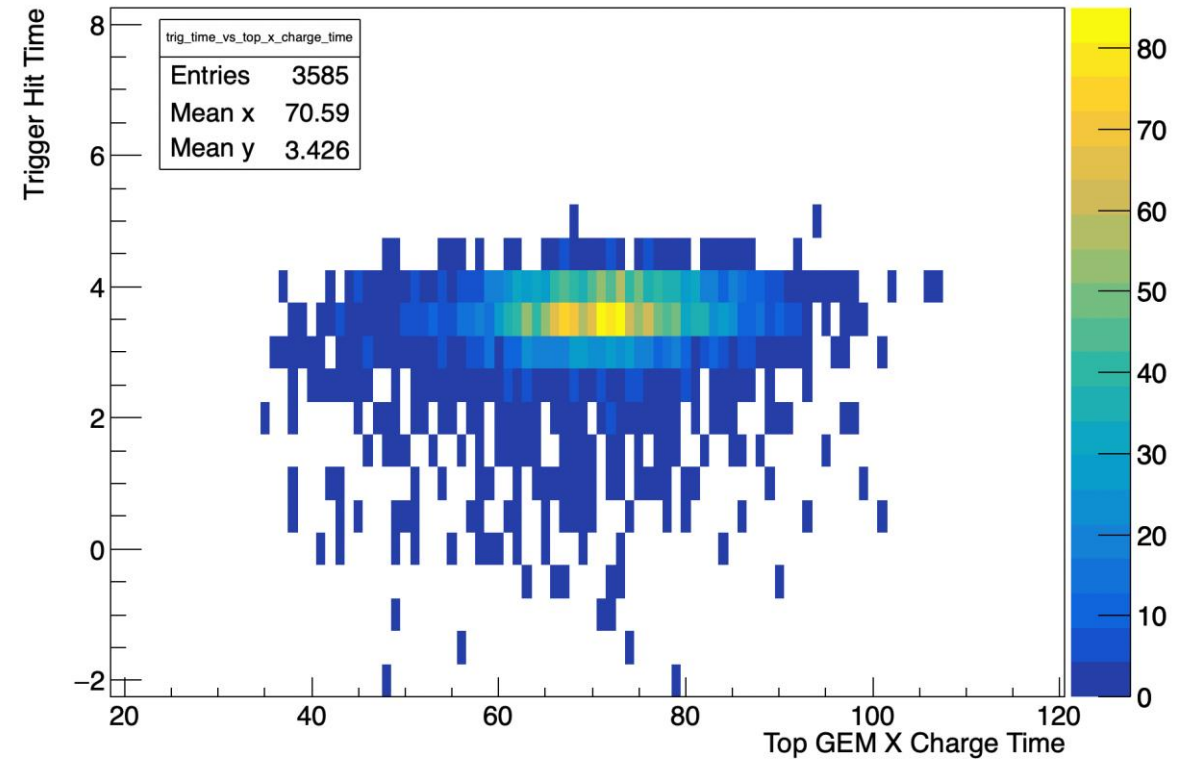
Trigger vs Left Top GEM X axis

Left: Trigger Hit Time vs Top GEM Cluster X Charge Time



All tracks

Left: Trigger Hit Time vs Top GEM Cluster X Charge Time

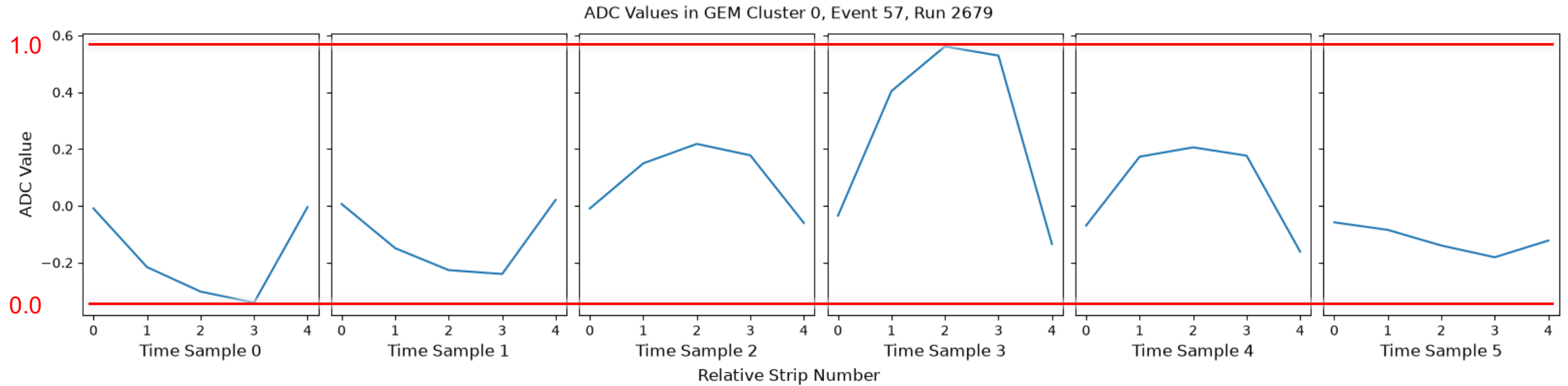


Single (one hit per detector per event) and
On-Paddles (hits scintillators) tracks

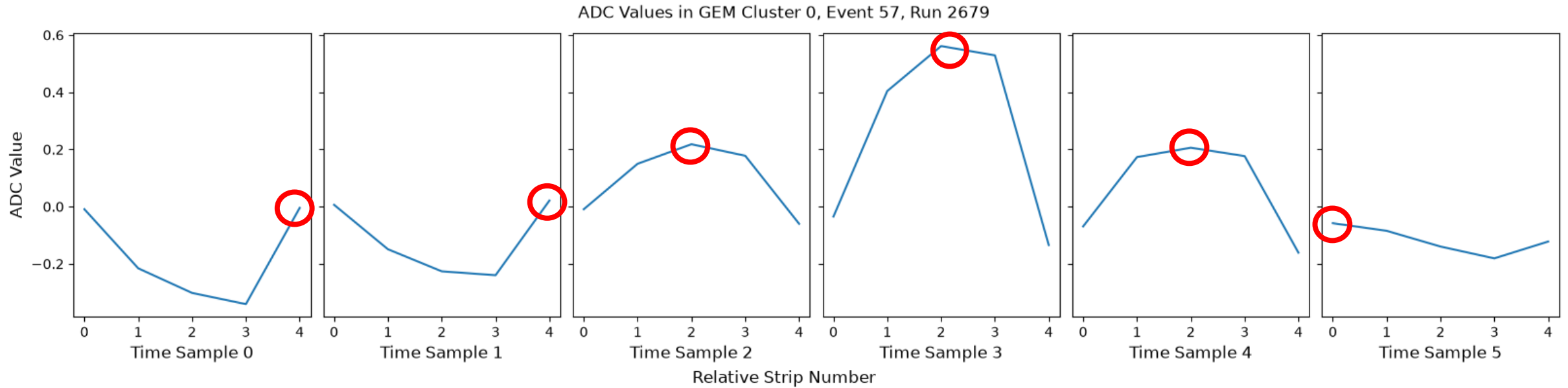
- Tested two rough methods of determining times of GEM clusters
 - No clear advantage between one method over the other
- Observed that many clusters identified by the GEMini code appeared noisy or difficult to interpret
- Clear correlation between cluster times in X and Y axis of individual GEMs
- Correlations between cluster times between GEMs and between GEMs and trigger hits is not clear

1. Implement improved pedestal subtraction code (Evan)
2. Revise and test cluster detection code
 - Study cluster signal shapes in channel/space domain
 - Improve detection of clusters of variable widths
 - Optimize for signal-to-noise ratio, efficiency of matching clusters with triggers
3. Redo this timing analysis
 - Improve method of calculating cluster time

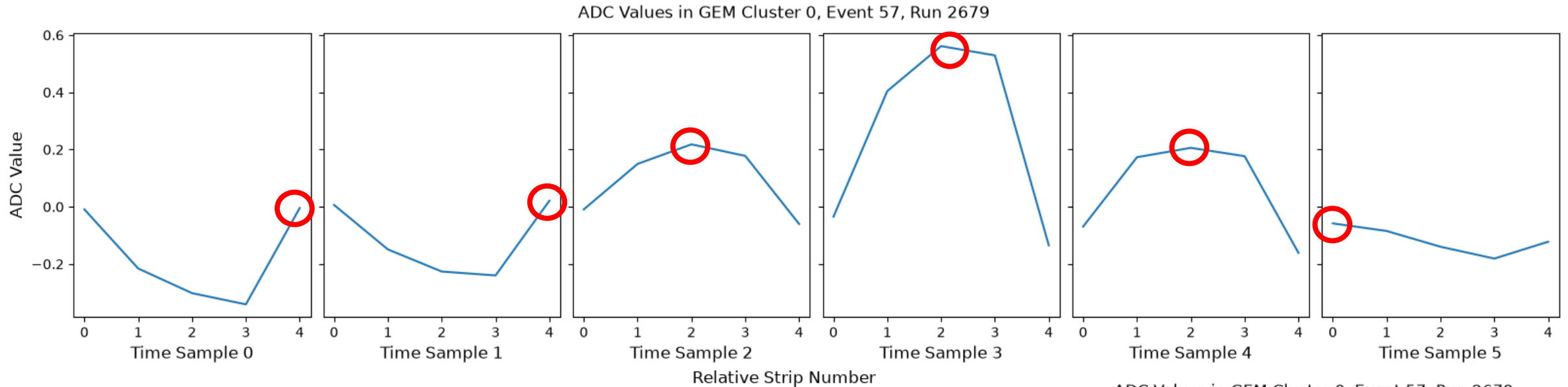
Thank you



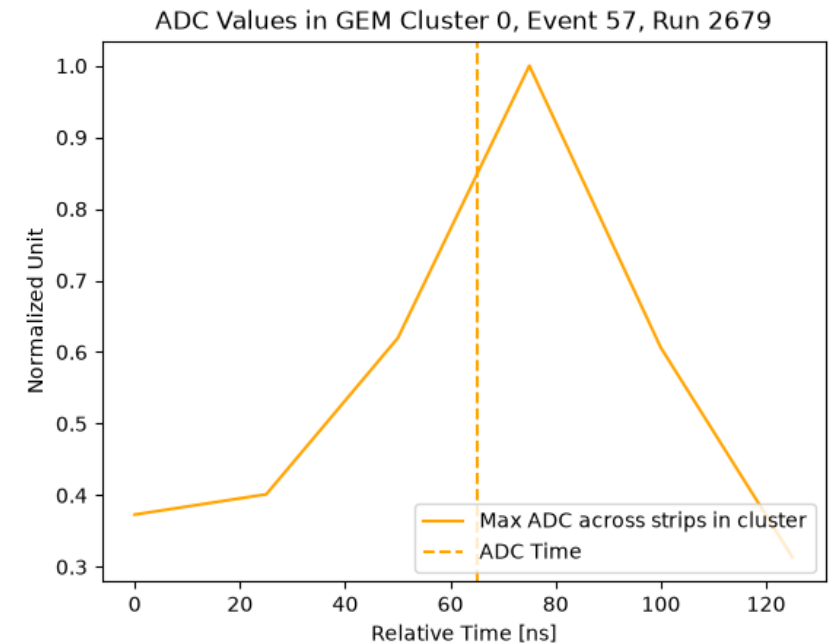
1. Normalize all values to the overall minimum and maximum

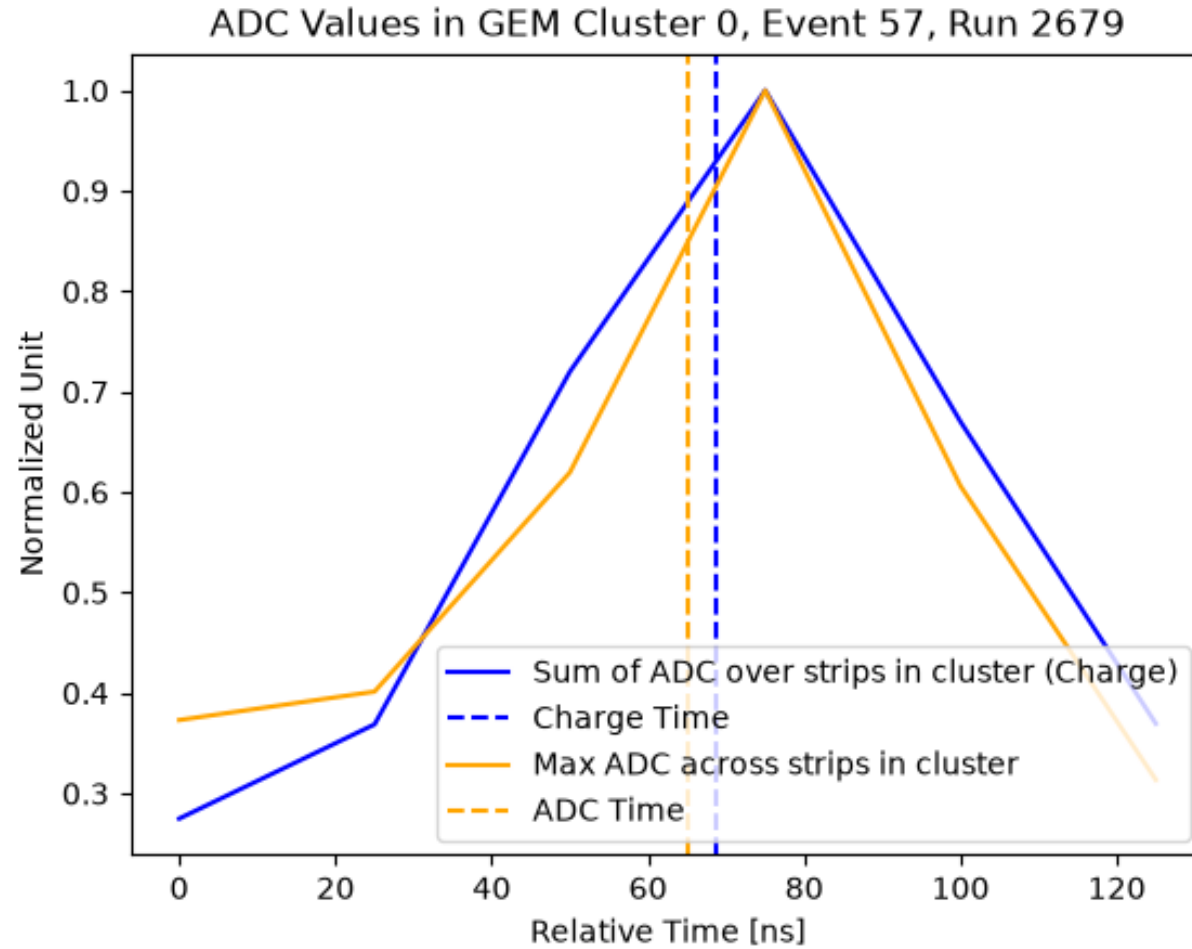


1. Normalize all values to the overall minimum and maximum
2. Find maximum channel value in each sample



1. Normalize all values to the overall minimum and maximum
2. Find maximum channel value in each sample
3. Calculate “mean” time weighted by these maximum values

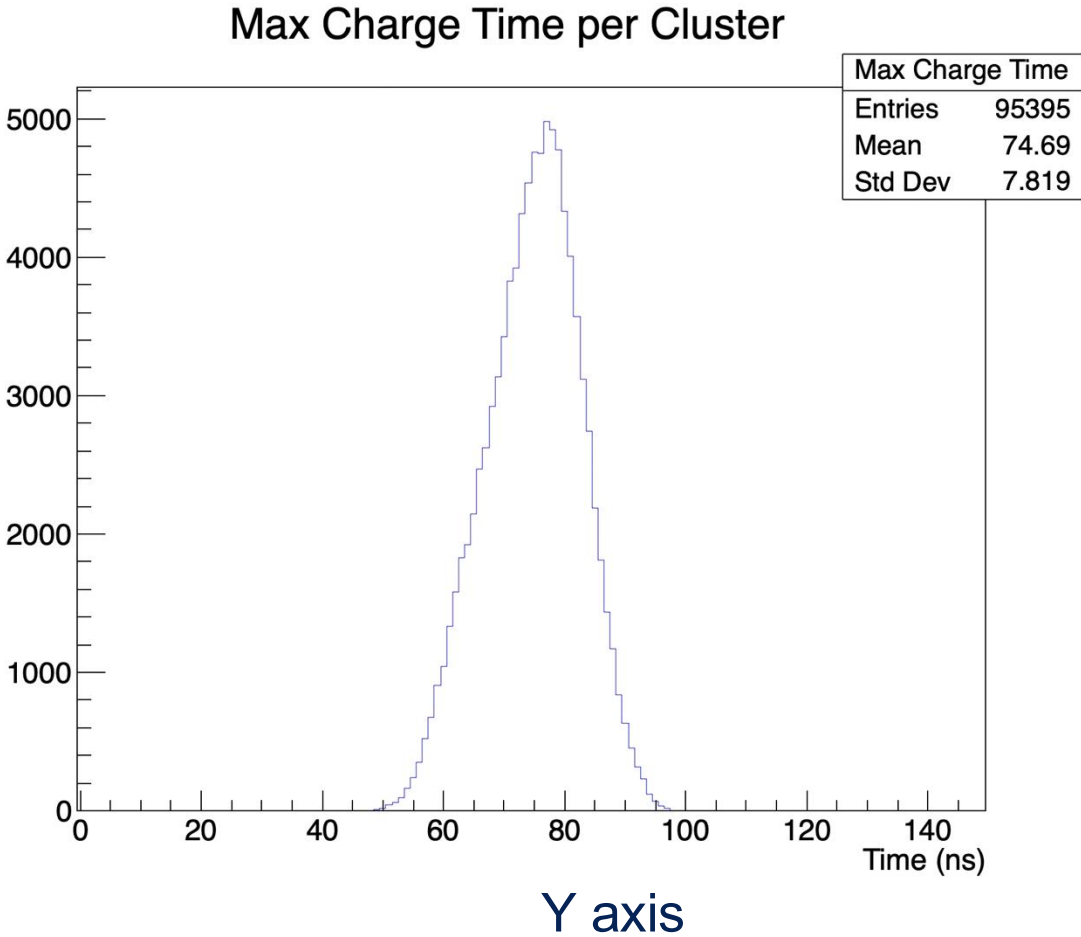
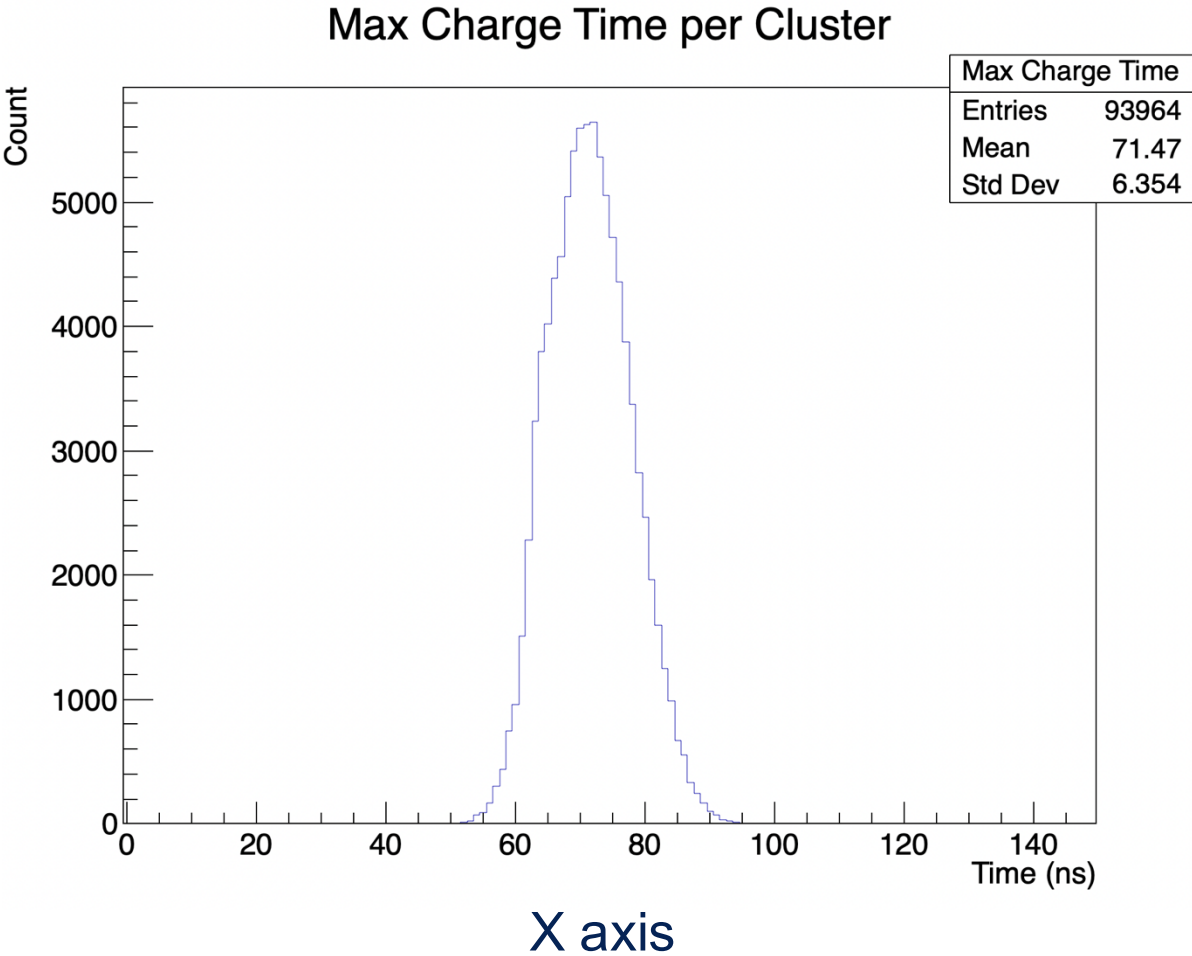




Charge Time

Left Top GEM clusters

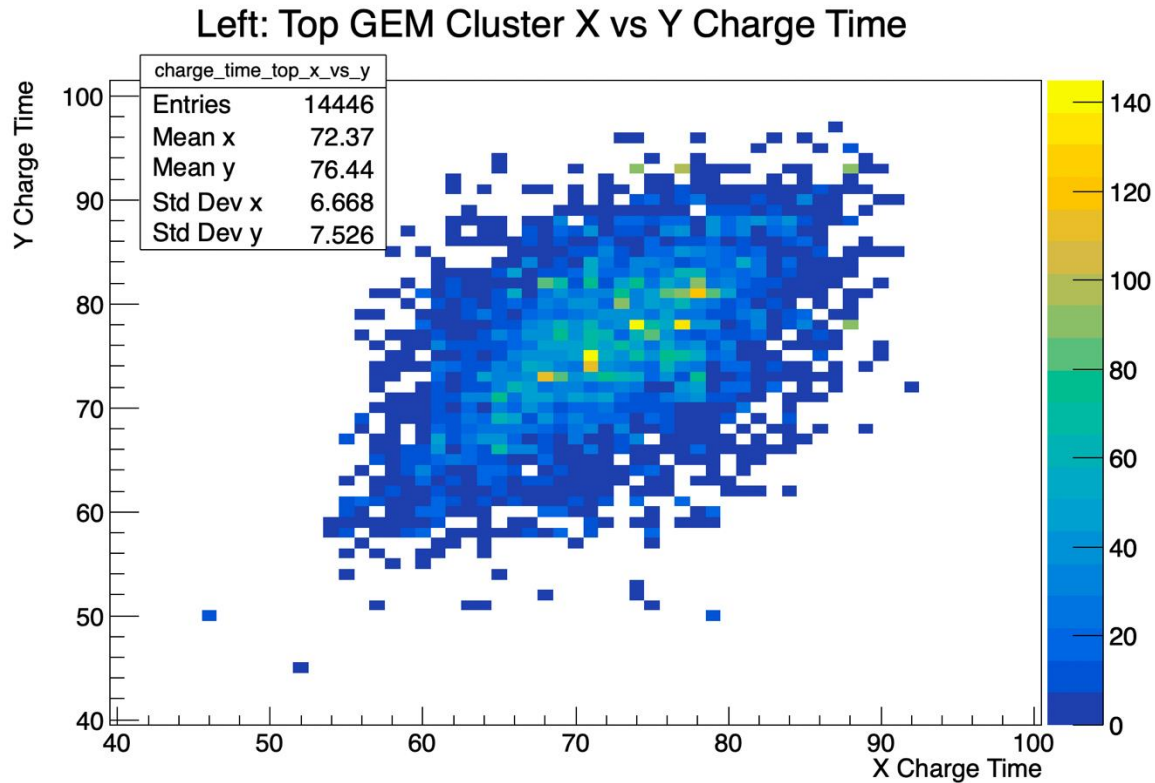
Run 3622 - Cosmic



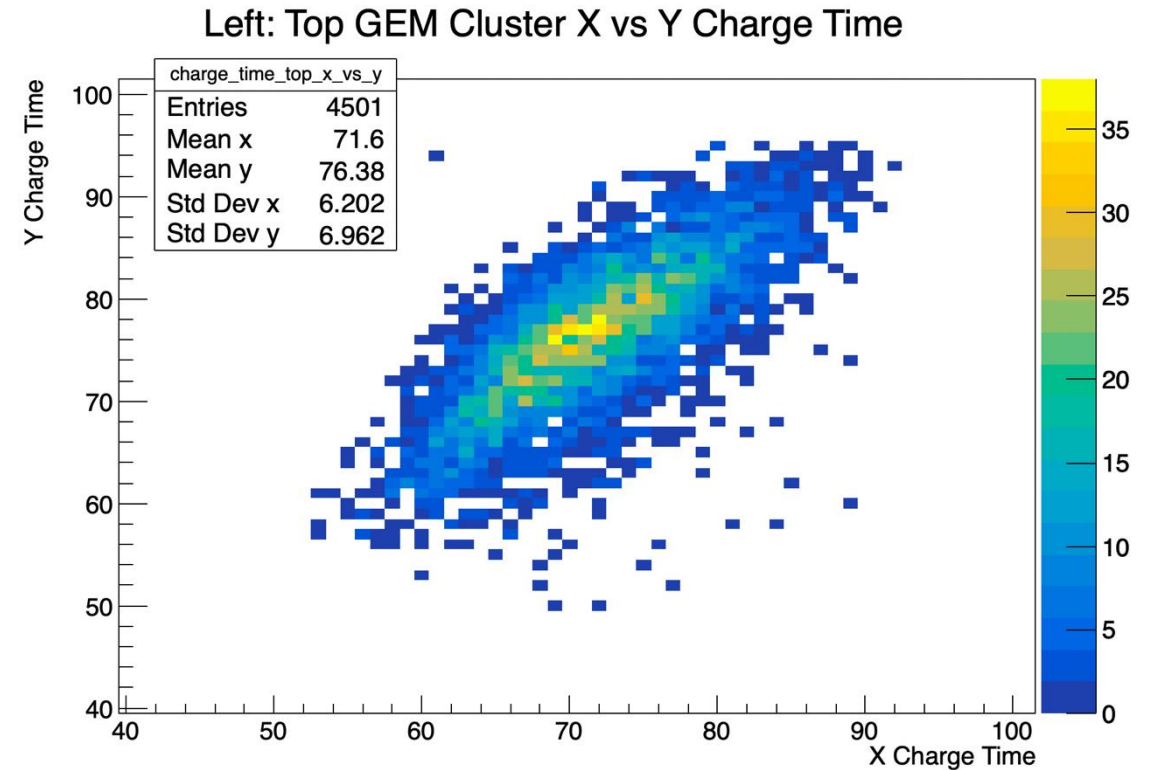
Charge Time

Left Top GEM clusters

Run 3622 - Cosmic



All tracks

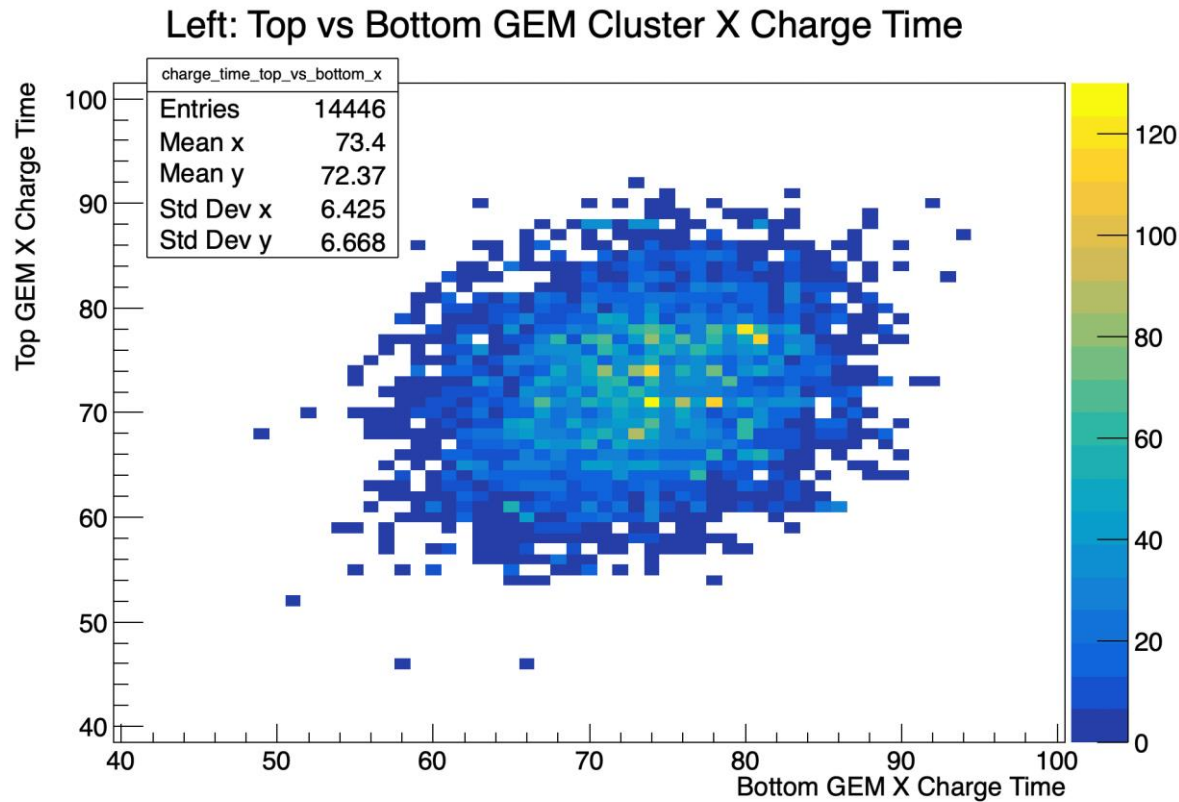


Single (one hit per detector per event) and
On-Paddle (hit scintillators) tracks

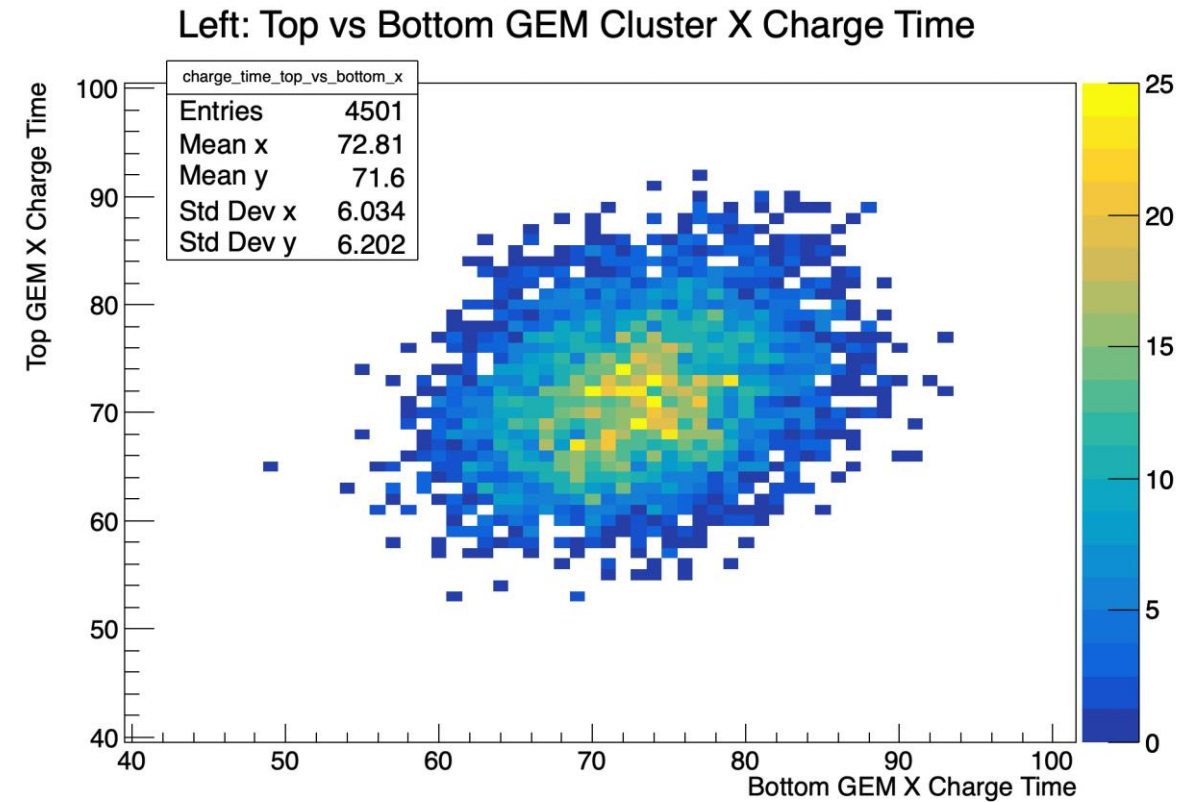
Charge Time

X axis clusters

Run 3622 - Cosmic



All tracks



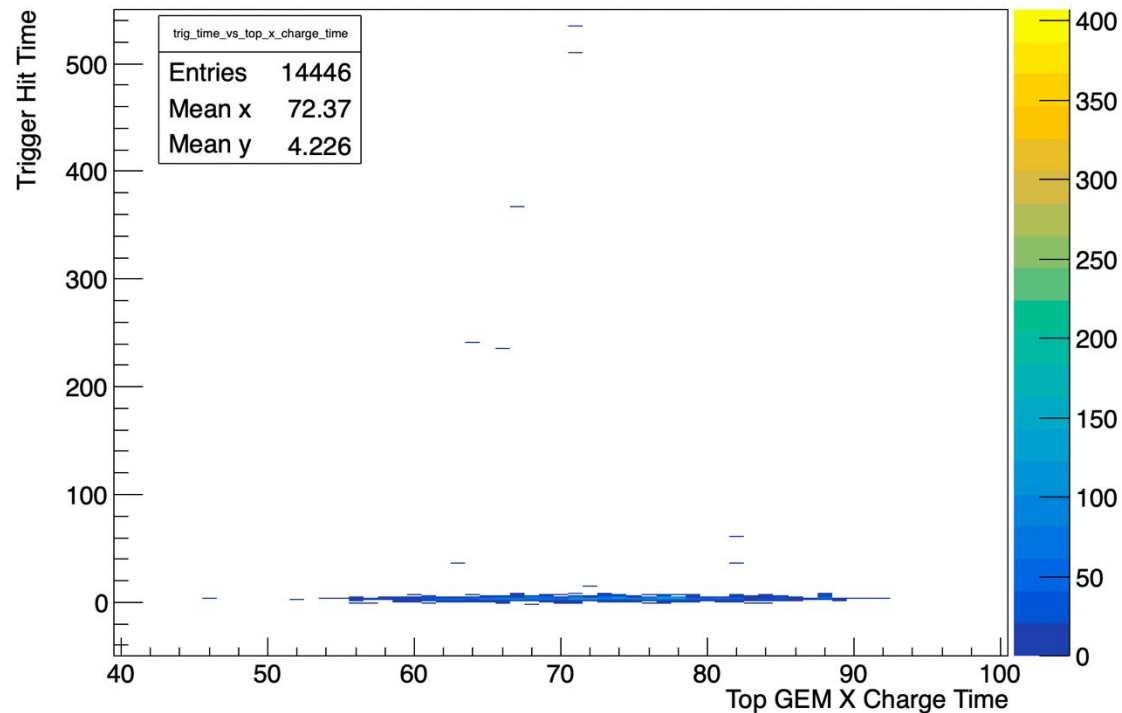
Single (one hit per detector per event) and
On-Paddle (hit scintillators) tracks

Charge Time

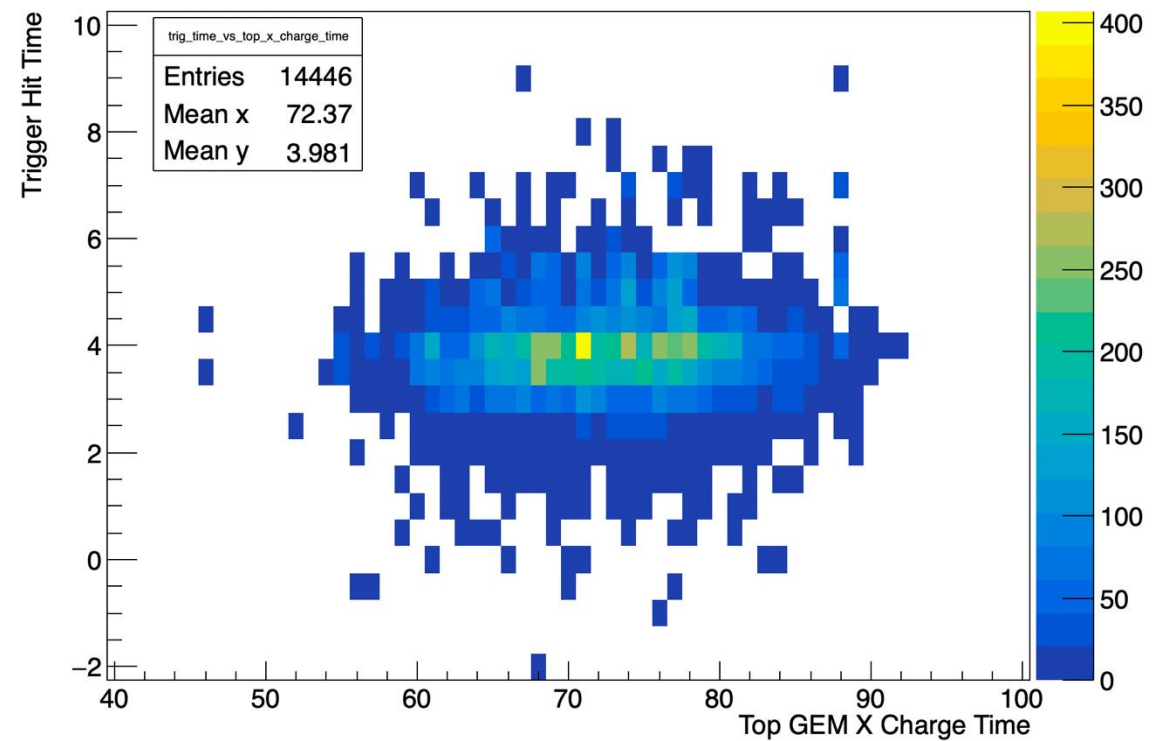
Run 3622 - Cosmic

Trigger vs Left Top GEM X axis

Left: Trigger Hit Time vs Top GEM Cluster X Charge Time



Left: Trigger Hit Time vs Top GEM Cluster X Charge Time



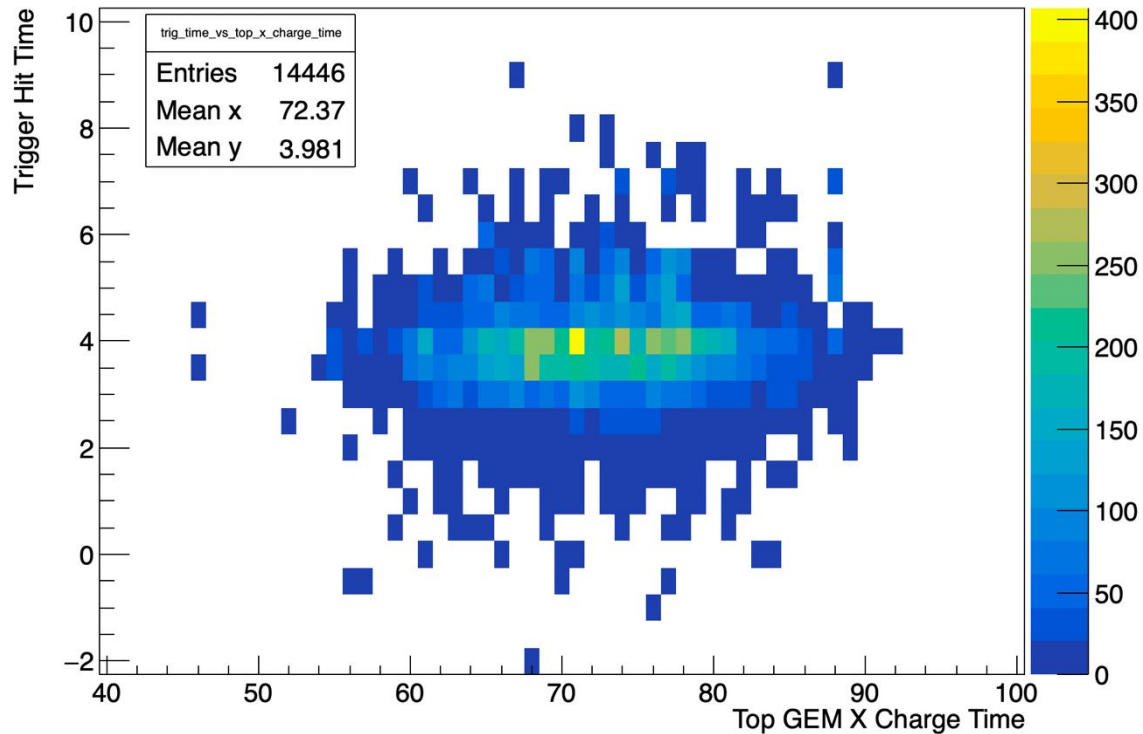
All tracks

Charge Time

Run 3622 - Cosmic

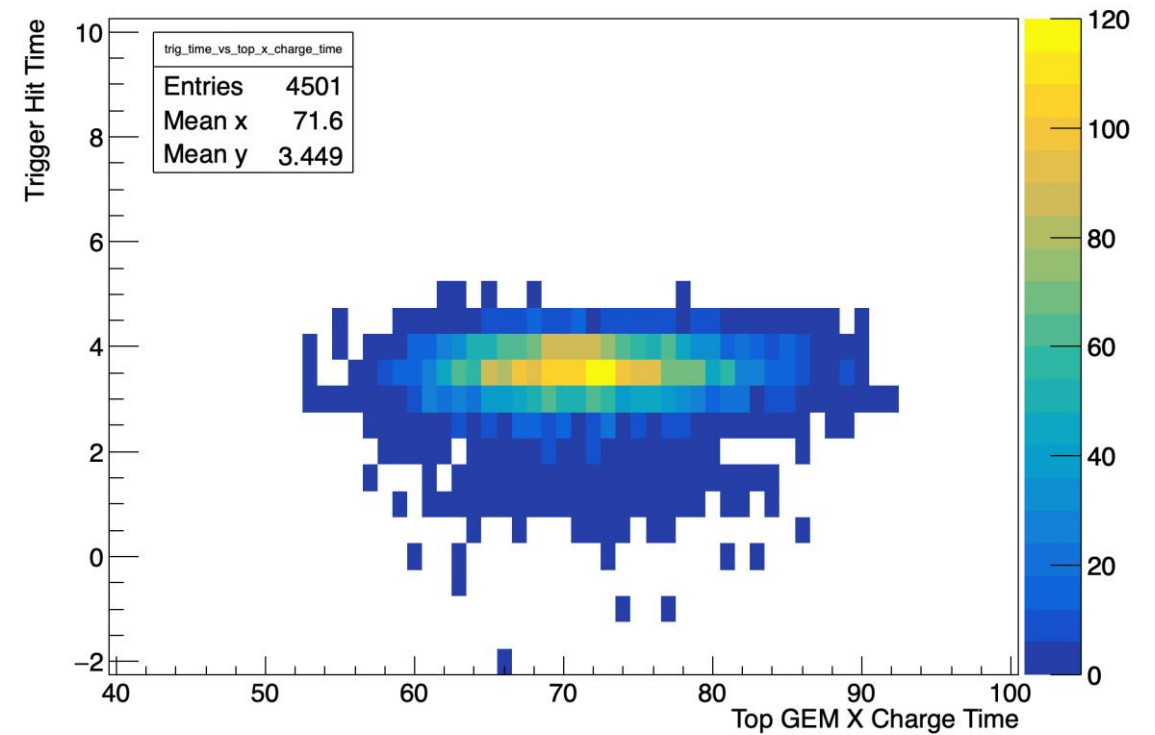
Trigger vs Left Top GEM X axis

Left: Trigger Hit Time vs Top GEM Cluster X Charge Time



All tracks

Left: Trigger Hit Time vs Top GEM Cluster X Charge Time



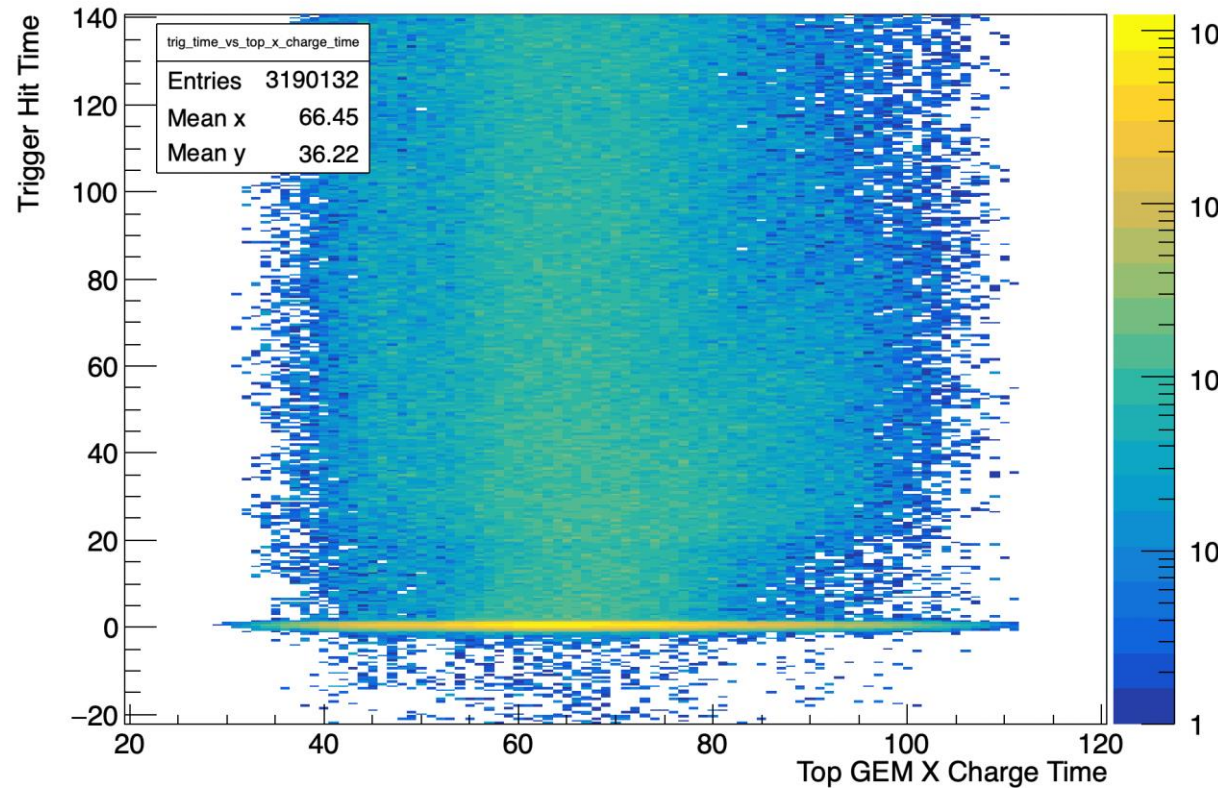
Single (one hit per detector per event) and
On-Paddles (hits scintillators) tracks

Charge Time

Run 4001 - Elastic

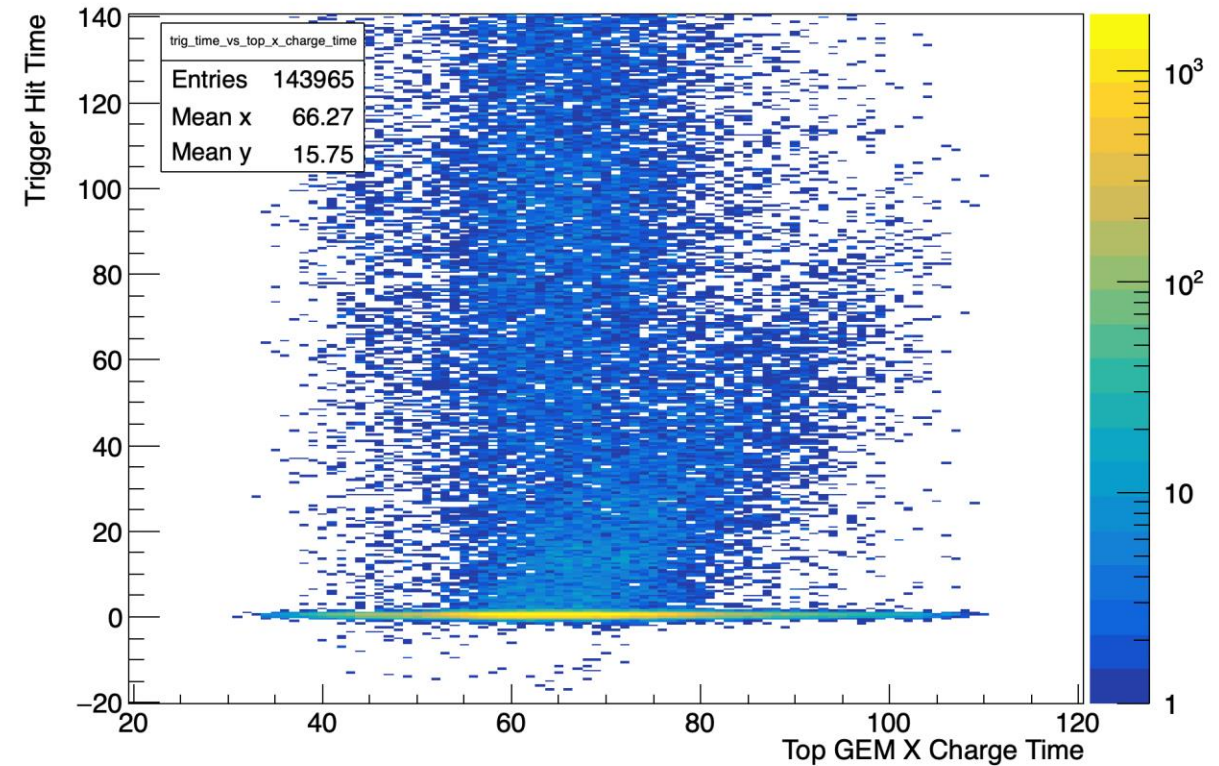
Trigger vs Right Top GEM X axis

Right: Trigger Hit Time vs Top GEM Cluster X Charge Time



All tracks

Right: Trigger Hit Time vs Top GEM Cluster X Charge Time



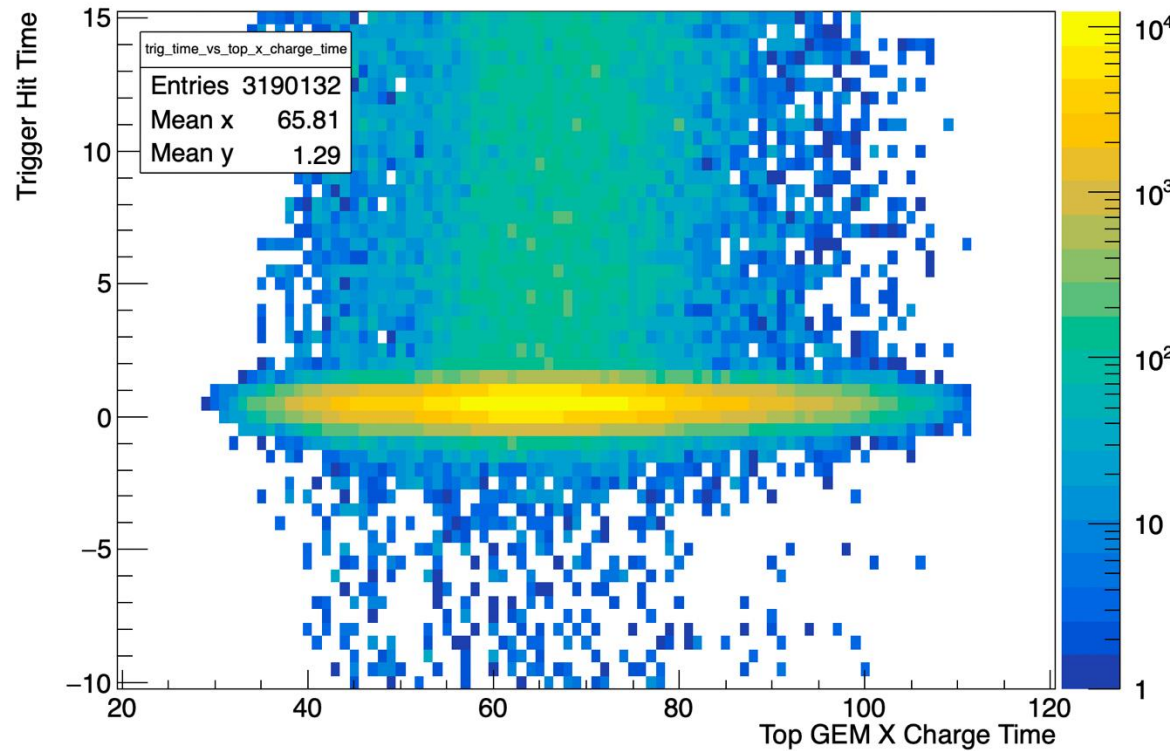
Best (closest to a trigger hit) and On-Paddles
(hits scintillators) tracks

Charge Time

Run 4001 - Elastic

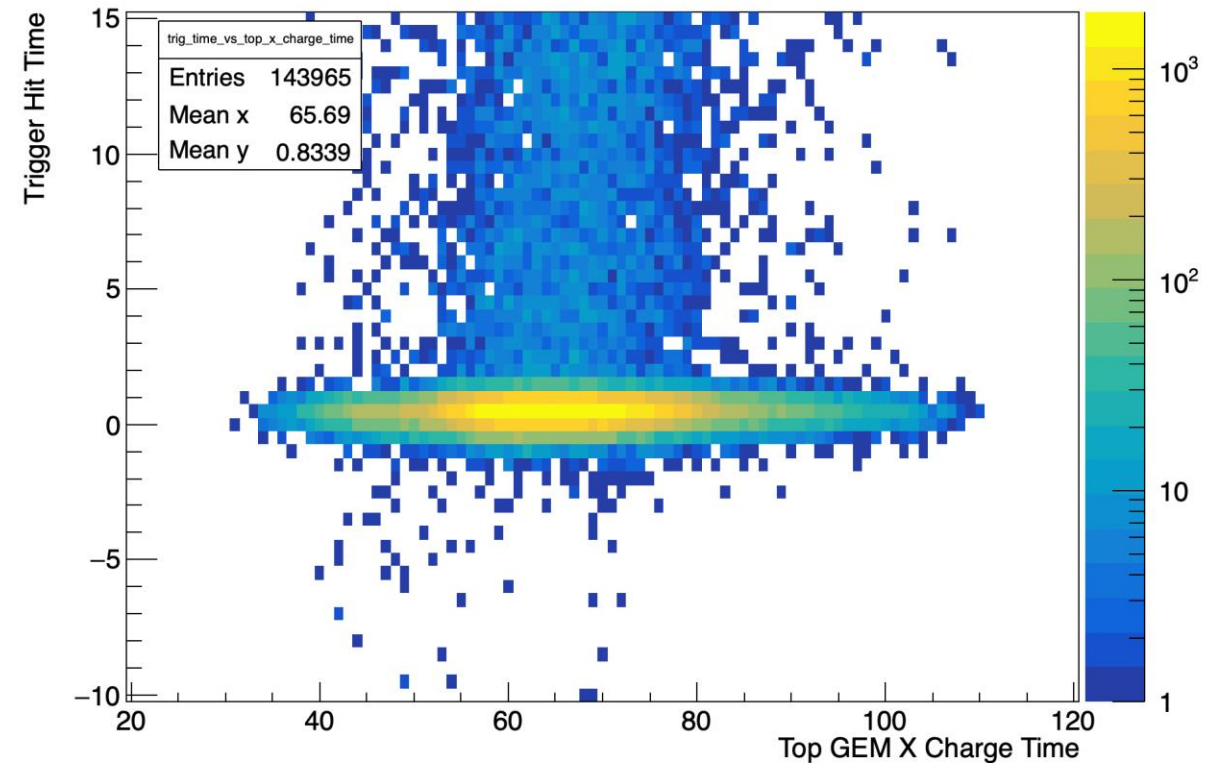
Trigger vs Right Top GEM X axis

Right: Trigger Hit Time vs Top GEM Cluster X Charge Time



All tracks

Right: Trigger Hit Time vs Top GEM Cluster X Charge Time



Best (closest to a trigger hit) and On-Paddles
(hits scintillators) tracks